# Gabor-Modulated Grouped Separable Convolutional Network for Hyperspectral Image Classification

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***Abstract*— Nowadays, convolutional neural network (CNN)-based deep learning (DL) models have been popularized in hyperspectral image classification (HSIC) and achieved significant accuracy gains, which is due to their hierarchical and nonlinear feature learning patterns. However, too deeper network structures may induce a huge amount of parameters and excessive computing overhead, leading to the need for plenty of labeled samples for training. Besides, highly abstract semantic features may not be the most suitable for hyperspectral land-cover classification tasks. To address these issues, we propose a fairly lightweight network model for HSIC, which is built on a type of exquisitely designed convolution module, namely, *grouped separable convolution* (GSC). Compared with the standard convolution, the designed GSC module combines grouped convolution with pointwise convolution, which not only greatly reduces the number of parameters of convolution kernels but also caters to the inherent 3-D cube style of hyperspectral image (HSI) data. Moreover, Gabor filters are introduced to modulate the GSC kernels, so as to further use relatively few convolution kernels with additional prior orientation and scale information for feature extraction. The experiments are carried out on four real hyperspectral datasets, and the experimental results reveal that the proposed model has low training cost and memory overhead. Compared with some existing deep network models that have been applied to HSIC, our proposed model can achieve competitive classification accuracy with fewer training samples.**

***Index Terms*— Gabor-modulated, grouped separable convolution (GSC), hyperspectral image classification (HSIC).**

## I. Introduction

HYPERSPECTRAL imaging spectrometer can capture hundreds or even thousands of narrow spectral bands from surface targets in a noncontact way, with spectral coverage ranges from visible light to near-infrared and short-wavelength infrared. Corresponding to the panchromatic and multispectral remote sensing images, hyperspectral images (HSIs) exhibit distinctive geometrical and statistical characteristics and have been extensively applied in land cover and use [1], precision agriculture [2], [3], mineral mapping [4], [5], national defense and military [6], and many other fields [7]. Among them, HSI classification (HSIC), which aims to assign a unique category label to each pixel, has been a hot research topic in the remote sensing community. However, due to the inherent nonlinear imaging process of HSIs, as well as the redundancy of spectral bands, the traditional classification methods still face with great challenges.

Generally speaking, the basic process of HSIC includes three parts. First, a series of preprocessing operations, such as data restoration [8], [9], denoising [10], [11], [12], and dimensionality reduction [13], [14], [15], are performed on the original HSI data, so as to obtain an enhanced input data representation. Then, feature extraction is performed on the input data. At present, the integration of spectral and spatial information is the mainstream strategy [16], [17]. Finally, the obtained feature maps are fed into a classifier to identify the category label of each pixel [18], [19], [20], and those classifiers used for HSIC are either constructed based on empirical risk minimization principle [e.g., $K$-nearest neighbor (KNN) [21], $K$-Means [22], and decision tree [23]], or designed on the basis of structural risk minimization principle [e.g., support vector machine (SVM) [24]].

With the promotion of deep learning (DL) technology, traditional handcrafted feature extraction methods are gradually turning to DL-based methods and exhibiting great potential in the HSIC scenario. It is because DL-based methods can automatically learn features from image data in a hierarchical and nonlinear pattern. In [25], single-layer autoencoder (AE) and multilayer stacked AEs (SAEs) were built to learn shallow and deep feature representations from hyperspectral data, respectively. Later, deep belief network (DBN) was also used to extract spectral–spatial integrated features [26]. Furthermore, convolutional neural networks (CNNs) were introduced to learn more efficient spatial and spectral feature representations from HSIs [27], [28]. Considering that hyperspectral data generally have the form of 3-D cube, it is beneficial to use 3-D convolution kernels directly. In [29], a 3-D CNN was constructed, and 3-D image patches were input into the network for deep feature extraction. Also in [30], 2-D CNN

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and 3-D CNN were combined to learn more diverse and abstract spatial feature representation.

The aforementioned DL models improved the classification accuracy to some extent. However, such performance gains primarily rely on sufficient training samples and exquisite structural design. On one hand, conventional deep CNN models generally have a large number of parameters to be learned; if the training samples are not enough, the model will easily fall into overfitting [31]. On the other hand, not similar to the natural image classification, which generally needs to extract more abstract semantic features that are closest to human understanding, so a very deep network is more beneficial, the recognition of ground objects in HSI mainly relies on spectral and spatial structural information, and some shallow features such as color, texture, and shape as well as some middle-layer attribute features are more desirable. With this regard, semantic features that are too deep and abstract may not be the best choice in HSIC tasks, and some medium-depth and relatively lightweight models are preferred. Among, Lin et al. [32] adopted $1 \times 1$ pointwise convolution to compress the feature maps into a lower dimensional space for reducing computing cost. Krizhevsky et al. [33] introduced grouped convolution and performed standard convolution operation on each group synchronously. Howard et al. [34] put forward a depthwise separable convolution (DWSC) and incorporated it into MobileNet model. Chollet substituted standard convolution in GoogLeNet's inception model [35] with DWSC [36]. Admittedly, although DWSC significantly reduces the number of parameters, it weakens the learning ability of convolution kernels, especially in terms of the correlation structure between channels. In addition, extremely separated convolution operation loses its adaptability to the inherent 3-D cube style of hyperspectral data; as a compromise, we have to stack more deeper separable convolution layers to fulfill satisfied feature representation. In comparison, grouped convolution, by designing a specific grouping scheme, can synchronously extract features within and between channels, thus obtaining stronger learning ability. Among, Liu et al. [37] proposed a lightweight shuffled group CNN (SG-CNN) and shown good performance in HSIC with limited samples. Mei et al. [38] introduced the grouped pixel embedding (GPE) module to leverage grouped convolution for local spectral–spatial feature extraction; considering that the vision transformer (ViT) module is used to extract global spatial features, the training cost of this model is very high.

Given no any prior knowledge, conventional CNNs have to use as many convolution kernels as possible to try to cover more comprehensive image feature representations, resulting in an imbalance between the large number of redundant parameters to be learned and the limited number of available training samples in the HSIC. With this in mind, some prior knowledge can be injected into the convolution kernels, so as to use less but more purposeful convolution kernels to learn discriminant features. Along this line, circular harmonics [39] were used to equip CNNs with both translation and rotation-equivariant kernels. However, the construction of rotation-equivariant kernel is very complex. On the contrary, Gabor filters provided joint time–frequency resolution, thus enhancing the capacity to extract low-level and middle-level features. When combining Gabor filters with conventional convolution kernels, not only shift invariance can be injected into the network but also rotation and scale invariance will be reinforced. In the early stages, Gabor filtered images were used as the input of CNNs [40]; among them, Yao et al. [41] used 2-D Gabor filters to generate three grayscale feature maps and then fed into CNNs. Subsequently, Gabor filters completely substituted the conventional convolution kernels and learned the corresponding parameters of Gabor kernels through back propagation optimization algorithm; among them, Sarwar et al. [42] adopted Gabor kernels with fixed parameters in the first two layers of CNNs, bringing a compromise between classification accuracy and training complexity. In both [43] and [44], a few early convolutional layers were replaced with parameterized Gabor convolutional layers, resulting in consistent boost in accuracy, robustness, and generalization test performance. However, due to the computational complexity of Gabor convolution operation and the corresponding parameter optimization procedure, its popularization is restricted. Specifically in the field of HSIC, the initial work came from the contributions of Chen et al. [45], who used Gabor filtering to extract edge and texture features, and then performed convolution on the Gabor filtered features. To combine Gabor representation and CNN architecture conveniently, Luan et al. [46] adopted 2-D Gabor filters to modulate standard convolution kernels and performed convolution operations with modulated Gabor orientation filters (GoFs), allowing the network to capture more robust features with regard to orientation and scale changes, while not bringing additional computational overhead. To further simplify the Gabor convolution operation, Liu et al. [47] introduced phase-induced Gabor filters for HSIC. Zhao et al. [48] designed a fractional Gabor convolution (FGC) module to extract features of multiscale, multiorientation, and semantic variation simultaneously. Jia et al. [49] used 3-D Gabor filtering to modulate the standard convolutional kernels, and then equipped with a well-designed multibranch architecture to adapt the 3-D Gabor-modulated kernels.

Inspired by the pioneering studies, we designed a novel grouped separable convolution (GSC) module and constructed a fairly lightweight convolutional network architecture (hereinafter dubbed "GS-GCN") for HSIC. The designed GSC combined grouped convolution with pointwise convolution, which not only greatly reduced the number of parameters of convolution kernels but also catered to the inherent 3-D cube style of HSI data. Moreover, Gabor filters with multiple orientations and scales were introduced to modulate the GSC kernels, so as to use few convolution kernels with additional prior orientation and scale information for feature extraction. The main innovative contributions of our work are threefolds.

1) First, the proposed GS-GCN model has a lightweight architecture; it is because the GSC module combines grouped convolution with pointwise convolution, which greatly reduces the number of parameters of convolution kernels. In the meantime, such grouping strategy can jointly learn the spectral–spatial integrated feature representation of HSIs.

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2) Second, we introduced Gabor-modulated convolution filters (GMFs) to substitute kernels in the GSC module, which aims to model rotation invariance and more complicated transformation invariance, thus enhancing deep feature representation with steerable orientation and scale capabilities.
3) Third, comprehensive experiments are carried out on four real hyperspectral data, and the experimental results reveal that the proposed GS-GCN model has low training cost and memory overhead. Compared with some existing deep network models that have been applied to HSIC, our proposed model can achieve competitive classification accuracy with fewer training samples.

The remainder of this article is organized as follows. Section II presents the architecture of the proposed GS-GCN model in a step-by-step fashion. Section III presents comprehensive experimental assessment of the GS-GCN model, which are carried out on four benchmark hyperspectral datasets. Section IV concludes this article with some remarks and hints at plausible future research lines.

## II. Proposed GS-GCN Model

This section provided a detailed description of our proposed GS-GCN model. First, we elaborated the overall framework of the GS-GCN model and its processing flow that was used for HSIC. Then, we presented detailed design of two key components in the GS-GCN architecture, including GSC and Gabor-modulated convolution module, respectively.

### A. Framework of GS-GCN Model

The overall framework of the GS-GCN model is shown in Fig. 1(d). It is known as an end-to-end feature learning and classification network, which mainly consists of four convolution blocks, one fully connected layer, and one softmax classifier layer. It should be noted that in the presented network configuration, each "GSGConv" layer denotes a GSC with GMFs, "BN" denotes batch normalization layer, "MP + R" denotes a composition of maximum pooling layer followed by ReLU activation, and "FC" denotes fully connected layer. In each "GSGConv" layer, GSCs are used to substitute standard convolutions, and then modulated by Gabor filters. For simplicity, we set the number of groups to two in each GSC, which will be discussed in Section III-B in detail. As for the GMFs, we adopted Gabor filters with different scales in different layers, while in each layer, Gabor filters with multiple orientations and fixed scale are adopted. In addition, performance analysis on different orientations of GMFs is also provided in Section III-C.

When the GS-GCN model is applied to HSIC, a batch of image cubes with size of $33 \times 33 \times \mathbf{d}$ are fed into the first GSGConv layer, in which the size of convolution kernel of GMFs is set to $3 \times 3$ and $1 \times 1$, corresponding to grouped convolution and pointwise convolution, respectively. Thus, the output feature maps of the first GSGConv have a size of $15 \times 15$. Likewise, after the second GSGConv layer, the size of output feature maps becomes $6 \times 6$, and after the third GSGConv layer, the size of output feature maps becomes $3 \times 3$ with padding 1. After the last GSGConv layer, the output feature maps are flattened into $1 \times 1$ and fed into the fully connected layer to obtain the final 1-D feature vectors, which will be used to calculate category scores via the loss function. Here, the Softmax cross-entropy loss function is adopted to deal with multiclass land cover.

To facilitate comparison, we also presented three related model frameworks, including 2-D standard-convolution-based CNN, namely, "CNN" [see Fig. 1(a)], 2-D standard convolution combined with GSC, namely, "GS-CNN" [see Fig. 1(b)], and 2-D GMFs with fixed orientations, namely, "Gabor convolutional network (GCN)" [see Fig. 1(c)]. As can be observed from Fig. 1(d), our GS-GCN combined GMFs with GSC and implemented at each layer.

### B. Architecture of Grouped Separable Convolution

In this section, we explained the design concept of GSC in detail. The GSC combines grouped convolution with $1 \times 1$ pointwise convolution, which greatly reduces the codependence between convolution kernels, as well as the number of parameters of the network.

*1) Grouped Convolution:* In a grouped convolution module, the channels of input and output feature maps are divided into the same number of groups, and standard convolutions are performed on each group. More specifically, for an input feature map $\mathbf{F}$ with shape of $H \times W \times C_{\text{in}}$, $\mathbf{F}$ is first split into $g$ nonoverlapped groups of feature maps along the channel dimension

$$\mathbf{F} = \{\mathbf{F}_1, \ldots, \mathbf{F}_i, \ldots, \mathbf{F}_g\} \tag{1}$$

where $g$ is the number of groups, and each grouped feature map $\mathbf{F}_i$ has a size of $H \times W \times (C_{\text{in}}/g)$. For each $\mathbf{F}_i$, we perform standard convolutions separately and obtain the output of each group. After that, the output features of all the groups are concatenated in sequence, as denoted by $\mathbf{F}'$

$$\mathbf{F}' = \text{Concat}(\mathbf{F}'_1, \ldots, \mathbf{F}'_i, \ldots, \mathbf{F}'_g). \tag{2}$$

Fig. 2 shows comparison of the process of standard convolution and grouped convolution, in which blue cubes stand for input and output feature maps, and yellow cubes represent standard convolution kernels. For the standard convolution [see Fig. 2(a)], supposing the input feature maps have a size of $H \times W \times C_{\text{in}}$, the corresponding output feature maps have a size of $H' \times W' \times C_{\text{out}}$, and $C_{\text{out}}$ number of standard convolution kernels with a size of $h \times w \times C_{\text{in}}$ are used for convolution operation. While for the grouped convolution [see Fig. 2(b)], to fulfill the same settings of the input and output feature maps, there will be the same $C_{\text{out}}$ number of standard convolution kernels, but with the size of $h \times w \times C_{\text{in}}/g$ and split into $g$ groups. As a result, the grouped convolution reduces the total number of convolution kernel parameters from $H' \times W' \times C_{\text{out}} \times C_{\text{in}}$ to $H' \times W' \times C_{\text{out}} \times C_{\text{in}}/g$, which means the number of parameters in a grouped convolution is $1/g$ the number of parameters in the corresponding standard convolution.

In a standard CNN model, the learned convolution kernels generally have plenty of redundancy; when all the

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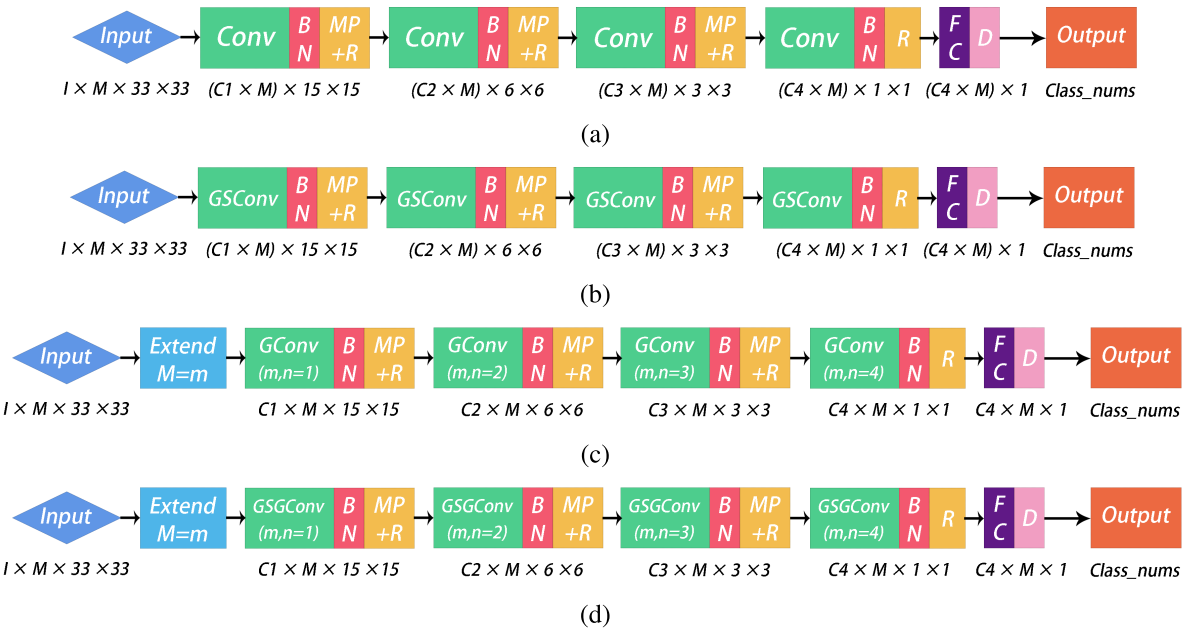



Fig. 1. Schematic of the proposed GS-GCN model. (a) 2-D standard-convolution-based CNN. (b) GSC. (c) 2-D standard convolution combined with GMFs with fixed orientations. (d) GS-GCN. In which "$I$" denotes the number of input channels, "$M$" refers to the number of orientations of Gabor filter, $C1$, $C2$, $C3$, and $C4$ are the number of channels in each layer, "BN" block denotes batch normalization layer, "MP" block denotes max-pooling layer, "$R$" denotes ReLU function, "FC" refers to fully connected layer, and "$D$" denotes dropout layer.

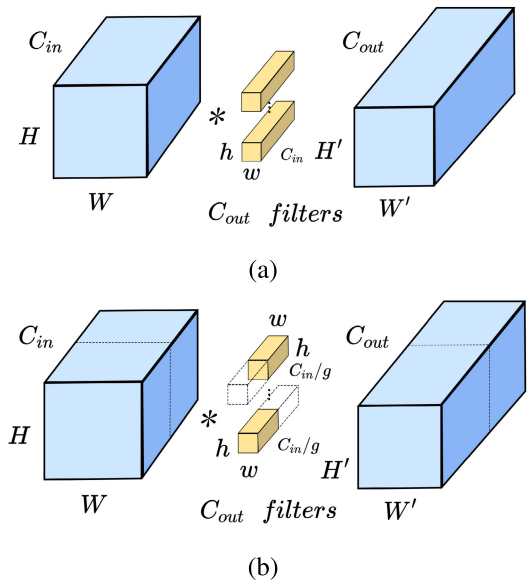



Fig. 2. (a) Standard convolution. (b) Grouped convolution with two number of groups.

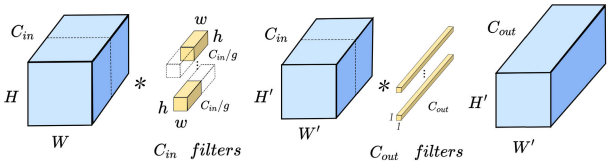



Fig. 3. GSC ($g = 2$).

feature channels are involved for convolution operation, it will inevitably bring further redundancy and noise interference due to the inherent dependence across channels. Relatively speaking, grouped convolution focuses on intragroup operations, and eliminates convolution of feature maps with other groups, and therefore, it can effectively reduce the codependence of convolution kernels across distinct groups.

*2) Grouped Separable Convolution:* Based on the grouped convolution, we further proposed the GSC. When compared with grouped convolution, GSC combines grouped convolution with pointwise convolution. As shown in Fig. 3, the convolution operation is further split into two steps: the first step is to reduce the size of feature maps by performing group convolution, which keeps the number of channels in the feature map, while the second step is to use pointwise convolution to expand the number of channels of feature map without changing the spatial scales.

To prove the superiority of GSC in the number of parameters, we set the size of input and output feature maps to be the same as before, and thus, the size of the convolution kernels in the GSC is $h \times w \times C_{\text{in}}/g$, and the size of the pointwise convolution kernel is $1 \times 1 \times C_{\text{in}}$. In this case, the parameters' quantity in standard convolution [see Fig. 2(a)] is $C_{\text{out}} \times C_{\text{in}} \times h \times w$, while the parameters' quantity in GSC is $C_{\text{in}} \times h \times w \times C_{\text{in}}/g + C_{\text{out}} \times C_{\text{in}} \times 1 \times 1$ during two stages. The ratio of parameters quantity of two types of convolution operations is shown below

$$\frac{P_g}{P_s} = \frac{C_{\text{in}}}{gC_{\text{out}}} + \frac{1}{hw} \tag{3}$$

where $P_g$ is the parameters' quantity of GSC, and $P_s$ refers to the parameters' quantity of standard convolution. As can be observed from (3), the number of groups, the ratio of the number of input and output channels, and the size of grouped convolution kernels will influence the total parameter

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volumes of GSC layer. When the number of groups equals the number of channels, the DWSC is established; in this case, each channel of the input feature map is treated as a unique group. Though the DWSC tends to have fewer parameters than GSC, it overlooks the correlation structure between feature channels and loses the ability to combine spectral and spatial information.

In a nutshell, GSC reduced the codependence between convolution kernels and network parameters simultaneously, which is more suitable for the characterization of essential 3-D structure of HSI. The subsequent experiments also show that GSC can maintain or even improve classification accuracy.

### C. Gabor Convolutional Network

GCN is a deep convolution network using Gabor-modulated convolution kernels (GMFs), which means the standard convolution kernels are substituted by GMFs.

*1) Gabor Filters:* Gabor filter is widely used to simulate the receptive field of simple cells in the visual cortex; as defined in the following equation, it is a windowed short-time Fourier transform combined with Gaussian window function:

$$G(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} e^{-\frac{1}{2}\left(\frac{x_r^2}{\sigma_x^2}+\frac{y_r^2}{\sigma_y^2}\right)+j\left(x\omega_x+y\omega_y+\Psi\right)} \tag{4}$$

where $xandy$ stand for the image coordinates, and $\sigma_x$ and $\sigma_y$ are the Gaussian standard deviations along the $x$- and $y$-axis, respectively. More specifically, $x_r = x\cos\theta + y\sin\theta$ and $y_r = y\cos\theta - x\sin\theta$, where $\theta$ is the orientation of Gabor filters. $\omega_x$ and $\omega_y$ are the frequency division of Gabor filters in the $x$- and $y$-axis, respectively, where $\omega_x = |\omega|\cos\theta$, $\omega_y = |\omega|\sin\theta$, $\Psi$ is the phase of Gabor filters, and $\omega$ and $\theta$ stand for the scale and orientation of Gabor filter, respectively.

In our work, the parameters of scale and orientation in Gabor filters are predefined, and $\theta \in [0, \pi]$ is set in 2/4/6/8 orientations uniformly, which generates 2/4/6/8 Gabor feature maps. Equally, the value range of $\omega$ can be selected in $(0, 1)$. In what follows, we noted the predefined Gabor filters with different scales and orientations as:

$$\phi = \left\{\phi_{m,n} | m \in \theta, n \in \omega\right\}. \tag{5}$$

*2) Gabor-Modulated Filters :* Supposing there are $M$ orientations and $N$ scales in a Gabor filter bank, to integrate steerable orientation properties into the standard learned convolution filters (hereinafter called "SLFs"), we keep multiple orientations filters with single-scale parameter in one layer, and then put the scale information into different convolution layers. The specific Gabor modulation strategy is shown in the following equation:

$$\begin{aligned} G_{i,m}^n &= C_{i,o} \circ \phi(m, n) \\ G_i^n &= \left\{G_{i,m}^n \middle| m \in \theta\right\} \end{aligned} \tag{6}$$

where $C_{i,o}$ stands for the $i$th input channel of an SLF. $\phi(m, n)$ refers to a set of Gabor filters with different scales and orientations, which has been defined in (5), and $\circ$ represents an element-by-element multiplication operation between $\phi(m, n)$ and each $C_{i,o}$. $G_i^n$ is the filter through modulation generated by $n$-scale Gabor filters.

Here, we used a visual illustration to further verify the performance of GMFs, and we presented parts of the first layer convolution kernels of the AlexNet model trained on the Pavia University dataset in Fig. 4(a), in which a darker color indicates a larger parameter value in the corresponding convolution kernel. Although Fig. 4(a) only reflects a rough distribution of convolution kernel parameters, we can still observe a certain degree of multiscale and multiorientation feature representation in some learned convolution kernels. This phenomenon is consistent with the modulation characteristics of Gabor filters and encourages us to directly modulate convolution kernels with multiscale and multiorientation feature extraction capabilities using fewer standard convolution kernels.

For comparison, Fig. 4(b) shows a set of Gabor filters under a fixed scale and various orientations, and Fig. 4(c) shows a series of Gabor-modulated filters. For those standard convolution kernels in Fig. 4(a), the filters within each blue box bear resemblance to the GMFs in Fig. 4(c), denoting that filters akin to GMFs can also be learned using standard convolution, but with plenty of redundancy. For example, the bottom-right group of convolution kernels marked by the red box in Fig. 4(a) appears to contain similar and redundant information; they all have a consistent overall distribution of convolution kernel parameters, characterized by intensity shifting from dark (top right) to light (bottom left). Such redundancy points to the inherent limitations of standard CNNs.

In the GSC schema, SLFs with the spatial scale of $h \times w$ are arranged into groups first, and then each grouped SLFs will be modulated by $M$-orientational Gabor filters to generate GMFs with $M$ times the number of SLFs. As shown in Fig. 5, the orientation parameter is set to $M = 4$, and supposing a group of SLFs has the size of $C \times h \times w$ (where $C$ stands for the number of channels). After modulation, we get a group of GMFs with the size of $M \times C \times h \times w$, resulting in enhanced feature representations without increasing learning parameters. As mentioned above, the generated GMFs are 3-D; if the size of SLFs is $C \times h \times w$, the intermediate parameter in a standard convolution kernel is $C_{\text{in}} \times C_{\text{out}} \times h \times w$, while the size of GMFs is $C_{\text{in}} \times C_{\text{out}} \times M \times h \times w$.

*3) GCN Convolution Operations:* The convolution procedure of GCN is the same as standard CNN. A output feature map $\widehat{F}$ can be represented as in the following equation:

$$\begin{aligned} \widehat{F} &= G\,\text{Conv}(F, G_i) \\ \widehat{F}_{i,j} &= \sum_{c=1}^{C} F^{(c)} \otimes G_{i,m=j}^{(c)} \end{aligned} \tag{7}$$

where $G_i$ refers to the $i$th GMF and $F$ stands for the input feature map [as shown in Fig. 5(b)]. $\widehat{F}_{i,j}$ refers to the $j$th orientational feature $\widehat{F}_i$, where $F^{(c)}$ and $G_{i,m}^{(c)}$ stand for $F$ and $G_{i,m}$ in the $c$th channel, respectively.

*4) Updating Strategy of GMFs:* In the constructed GS-GCN framework, the weight of GMFs is calculated in a forward propagation. During back propagation, we only need to update parameters of SLFs, while for the parameters of orientations and scales of Gabor filters, we only need to add gradients of

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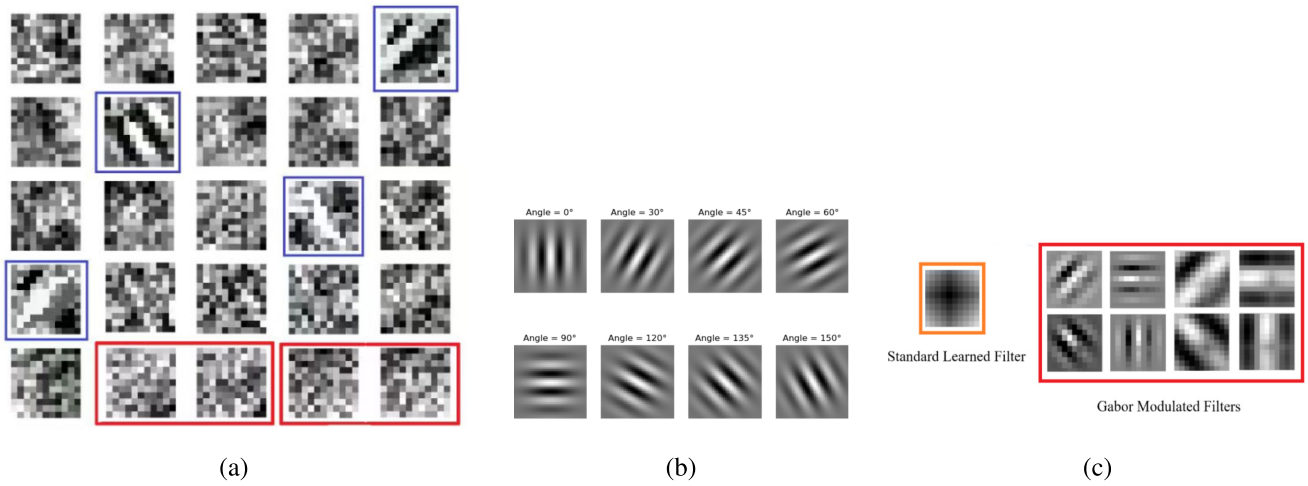



Fig. 4. Kernel visualization and comparison.

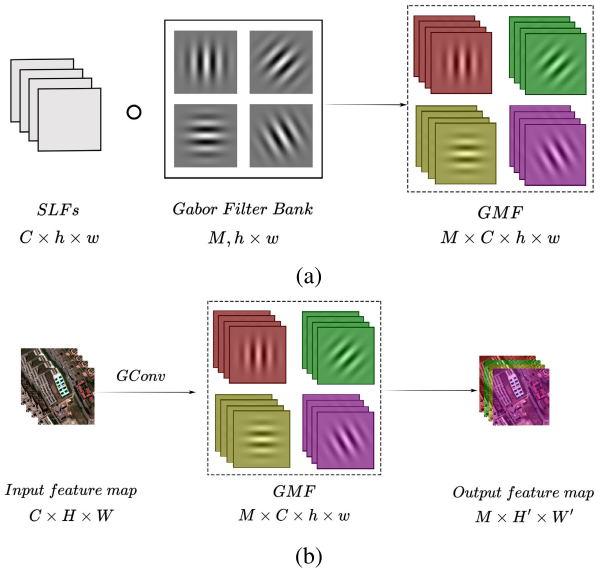



Fig. 5. Modulation and convolution process of $M$-orientation GMFs. (a) Modulation process of GMFs. (b) Convolution process of GCN with $M$-orientation GMFs.

the subfilters in GMFs to the corresponding SLFs. Specifically, the updating process is described as in the following equation:

$$\delta = \sum_{m=1}^{M} \frac{\partial L}{\partial G_{i,m}} \circ \phi(m,n) = \frac{\partial L}{\partial C_{i,o}}$$
$$C_{i,o} = C_{i,o} - \eta\delta \tag{8}$$

where $L$ refers to the loss function, $\delta$ is the gradient, and $\eta$ is the learning rate.

### D. Gabor-Modulated Grouped Separable Convolutional Network

Our GS-GCN model combines the GSC with GCN. Supposing the size of input feature maps is $C_{\text{in}} \times H \times W$ and the size of output feature maps to be $C_{\text{out}} \times H' \times W'$, the specific convolution process is as follows.

First, the channels of input feature map are expanded by a factor of $M$ to accommodate the convolution operation with GMFs, where $M$ stands for the orientations of GMFs. Then, GMFs are applied to the channel-expanded input data; in this case, $C'_{\text{in}} = M \times C_{\text{in}}$.

Second, GSC are conducted with two sequence steps. In the first stage, grouped convolution is adopted to reduce the spatial size of the feature map while keeping its channel number unchanged, for example, the input data with size of $M \times C_{\text{in}} \times H \times W$ will be changed into $M \times C_{\text{in}} \times H' \times W'$ after the grouped convolution, where $H' < H$ and $W' < W$. Fig. 6 presents three grouping strategies, which divide the input feature maps into 1/2/4 groups, respectively, according to the number of channels $C_{\text{in}}$. Here, we take $g = 4$ as an example; in grouped convolution, each group performs convolution operation independently. Considering that GMFs are a combination of SLFs modulated in multiple orientations, the corresponding convolution process becomes 3-D. For example, in the first group, the input feature map is divided into 3-D cubes with the size of $M \times H \times W$ and a total of $C_{\text{in}}/4$ feature cubes. So the cube size of the output feature map should be $M \times H' \times W'$ with a total of $C_{\text{in}}/4$. In this case, we need $(C_{\text{in}}/4) \times (C_{\text{in}}/4)$ number of GMFs and each $m$-orientation GMF is composed of a group of SLFs modulated in $M$ number of orientations. In the second stage, pointwise convolution is applied to expand the number of channels without changing the spatial scale of feature maps.

## III. Experimental Results and Discussion

This section is organized as follows. Section III-A describes four hyperspectral datasets that have been used for assessing the GS-GCN model, including Indian Pines, Salinas, University of Pavia, and Fanglu Tea Farm. Section III-B shows comparison of the performance of different grouped schemes and applying GSCs layer-by-layer to obtain a high-performance GSC scheme. Section III-C analyzes the performance of different numbers of channels and orientations of Gabor filters with four compared models in Fig. 1. Section III-D shows comparison of the classification performance of our model

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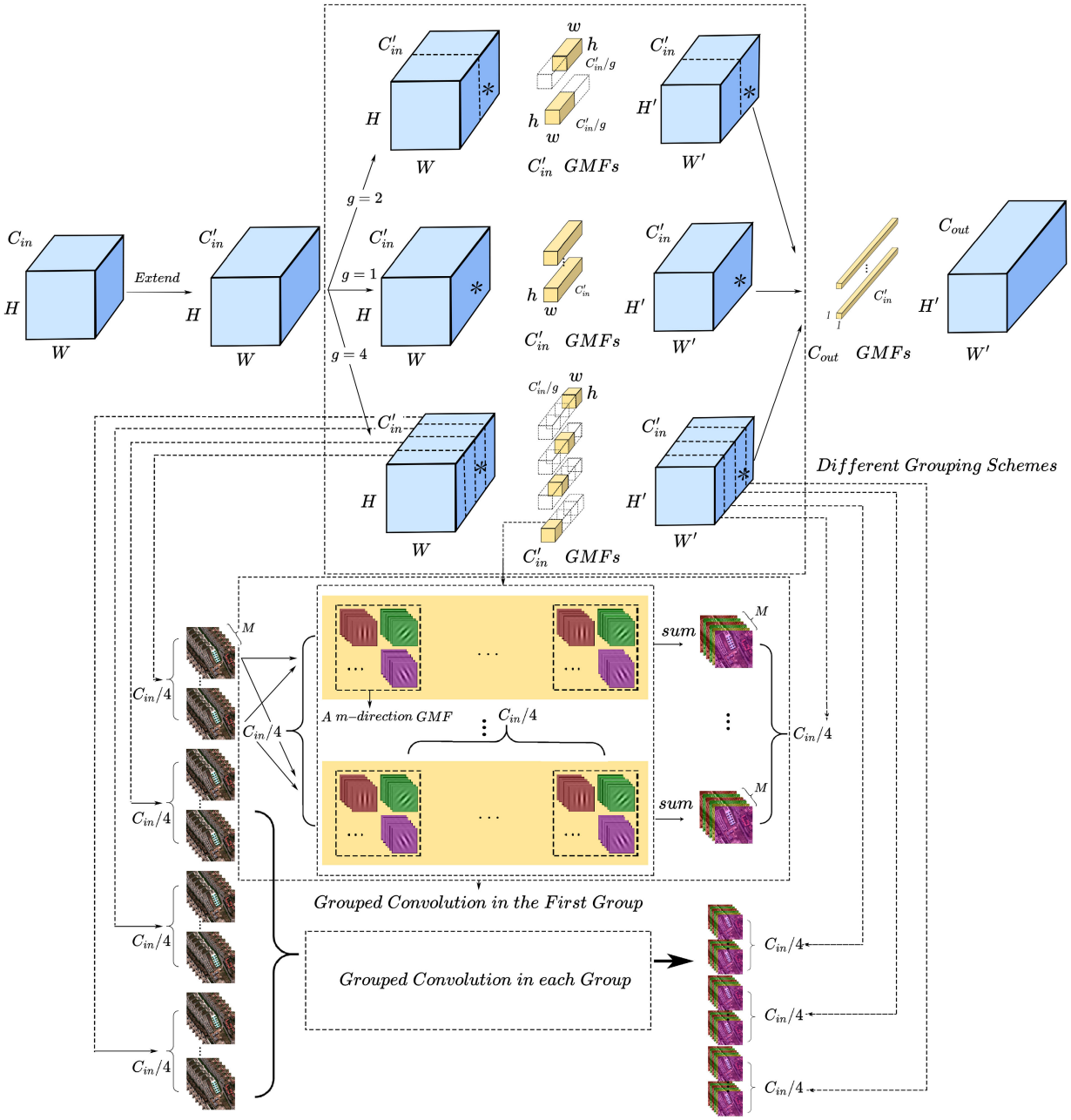



Fig. 6. Size of input feature map is $H \times W \times C_{\text{in}}$ and the size of output feature map is $H' \times W' \times C_{\text{out}}$. We set group number $g = 1, 2, 4$ for example. For $m$-orientation GMFs, we need to expand the input channels from $C_{\text{in}}$ to $C'_{\text{in}}$, where $C'_{\text{in}} = \text{MC}_{\text{in}}$.

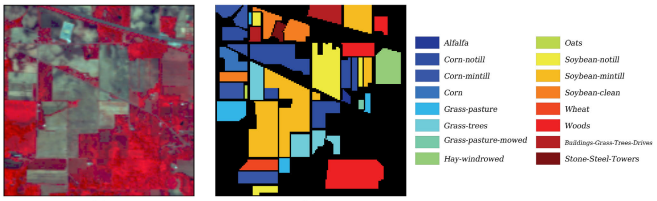
with other popular network models. Section III-E carries out experimental analysis on different proportion of training sample sets. Finally, Section III-F analyzes the computational complexity of the GS-GCN model.

As a prelude, we first give parameter settings in our experiments. First, principal component analysis (PCA) is performed on the original HSIs for dimensional reduction. Concretely, the spectral dimension of the Indian Pines and Salinas datasets is down to 20, and the spectral dimension of the Pavia University and Fanglu Tea Farm datasets is down to 10. Then, the cube-style training samples are cut from the dataset with a spatial patch size of 33 × 33 in our experiments.

Aiming to the split of training and test set, for Indian Pines and Salinas, we take 50 samples of each class as training samples, and the remaining samples for testing, while for Pavia University and Fanglu Tea Farm, we take 100 samples of each class as training samples. In addition, if the total number of samples in a class is less than the specified size, 80% of the samples in this class will be used for training, and the remaining 20% are used for testing. The training and test samples are randomly scrambled, and the batch size of training and testing is set to 128.

In the training process, we use Adam optimizer to update the parameters for 50 training epochs, the learning rate is $\eta = 0.01$, and the weight decay coefficient is $3e$-05. To reduce

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(a) (b)

Fig. 7. (a) False-color composition of the AVIRIS Indian Pines scene. (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

TABLE I

NUMBER OF TRAINING, TESTING, AND TOTAL SAMPLES FOR EACH LAND-COVER CLASS IN THE INDIAN PINES DATASET

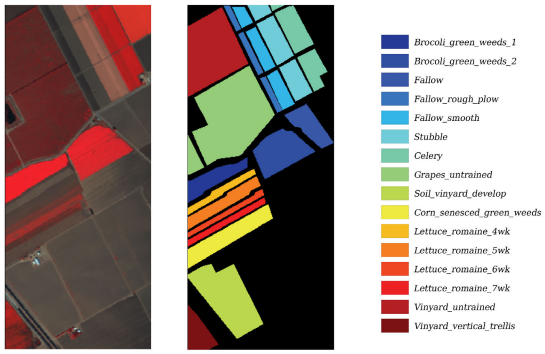
| Class No. | Land Cover Type | Train Num | Test Num | Total Num |
|---|---|---|---|---|
| C1 | Alfalfa | 36 | 10 | 46 |
| C2 | Corn-notill | 50 | 1378 | 1428 |
| C3 | Corn-mintill | 50 | 780 | 830 |
| C4 | Corn | 50 | 187 | 237 |
| C5 | Grass-pasture | 50 | 433 | 483 |
| C6 | Grass-trees | 50 | 680 | 730 |
| C7 | Grass-pasture-mowed | 22 | 6 | 28 |
| C8 | Hay-windrowed | 50 | 428 | 478 |
| C9 | Oats | 16 | 4 | 20 |
| C10 | Soybean-notill | 50 | 922 | 972 |
| C11 | Soybean-mintill | 50 | 2405 | 2455 |
| C12 | Soybean-clean | 50 | 543 | 593 |
| C13 | Wheat | 50 | 155 | 205 |
| C14 | Woods | 50 | 1215 | 1265 |
| C15 | Buildings-Grass-Trees-Drives | 50 | 336 | 386 |
| C16 | Stone-Steel-Towers | 50 | 43 | 93 |
| | Total | 724 | 9525 | 10249 |

the uncertainty caused by randomness and ensure reliability of the results, we repeat the experiments ten times. Finally, we would like to emphasize that all our experiments are conducted on Python 3.6 in a desktop PC equipped with an NVIDIA GetForce RTX 2080 Ti and 32 GB of RAM.

### A. Hyperspectral Datasets

In our experiments, four benchmark hyperspectral datasets are used to evaluate the performance of our proposed GS-GCN model.

*1) Indian Pines:* The Indian Pines was collected by the NASA AVIRIS instrument over the Indian Pines region in Northwestern Indiana in 1992. This dataset consists of 145 × 145 pixels. The imaging wavelength range of AVIRIS imaging spectrometer is 0.4–2.5 $\mu$m. The ground objects are continuously imaged in 220 consecutive bands, but 20 bands cannot be reflected by water. Therefore, we generally use the remaining 200 bands after removing these 20 bands as the research object. For the purpose of classification, 16 mutually exclusive ground-truth classes are provided. This scene shows the highly mixed pixels in all available classes and the unbalanced number of tag pixels in each class. For illustrative purposes, Fig. 7(a) shows a false-color composite of the image, while Fig. 7(b) shows the corresponding ground-truth map. There are 10 249 labeled samples from 16 different land-cover types in this dataset, whose training and testing sets have been comprehensively provided in Table I.




(a) (b)

Fig. 8. (a) False-color composition of the AVIRIS Salinas scene. (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

TABLE II

NUMBER OF TRAINING, TESTING, AND TOTAL SAMPLES FOR EACH LAND-COVER TYPE IN THE SALINAS DATASET

| Class No. | Land Cover Type | Train Num | Test Num | Total Num |
|---|---|---|---|---|
| C1 | Brocoli-green-weeds-1 | 50 | 1959 | 2009 |
| C2 | Brocoli-green-weeds-2 | 50 | 3676 | 3726 |
| C3 | Fallow | 50 | 1926 | 1976 |
| C4 | Fallow-rough-plow | 50 | 1344 | 1394 |
| C5 | Fallow-smooth | 50 | 2628 | 2678 |
| C6 | Stubble | 50 | 3909 | 3959 |
| C7 | Celery | 50 | 3529 | 3579 |
| C8 | Grapes-untrained | 50 | 11221 | 11271 |
| C9 | Soil-vinyard-develop | 50 | 6153 | 6203 |
| C10 | Corn-senesced-green-weeds | 50 | 3228 | 3278 |
| C11 | Lettuce-romaine-4wk | 50 | 1018 | 1068 |
| C12 | Lettuce-romaine-5wk | 50 | 1877 | 1927 |
| C13 | Lettuce-romaine-6wk | 50 | 866 | 916 |
| C14 | Lettuce-romaine-7wk | 50 | 1020 | 1070 |
| C15 | Vinyard-untrained | 50 | 7218 | 7268 |
| C16 | Vinyard-vertical-trellis | 50 | 1757 | 1807 |
| | Total | 800 | 53329 | 54129 |

*2) Salinas:* The Salinas data were also taken by AVIRIS imaging spectrometer in Salinas Valley, CA, USA. Its spatial resolution is up to 3.7 m, with a spatial size of 512 × 217. Similarly, this dataset also excluded 20 wavebands that cannot be reflected by water, resulting in a total of 204 wavebands remained. Fig. 8 presents a false-color composite of the image and the corresponding ground-truth map. A total number of 54 129 samples from 16 crop categories are labeled, of which the training and testing sample numbers are distinctly displayed in Table II.

*3) University of Pavia:* The University of Pavia dataset was acquired by the ROSIS sensor on July 8, 2002. The spatial scale of these data is 610 × 340 pixels, 1.3-m spatial resolution, and 103 spectral bands ranging from 0.43 to 0.86 $\mu$m. Nine practical ground classifications of interest are provided, including urban characteristics, soil, and vegetation. Fig. 9 shows a false-color composite of the image and the corresponding ground-truth picture. The dataset has 42 776 labeled samples altogether. Table III presents the number of training and testing samples of different interest classes sufficiently used in the experiment.

*4) Fanglu Tea Farm:* The Fanglu Tea Farm dataset was acquired by the Pushbroom Hyperspectral Imager (PHI) in

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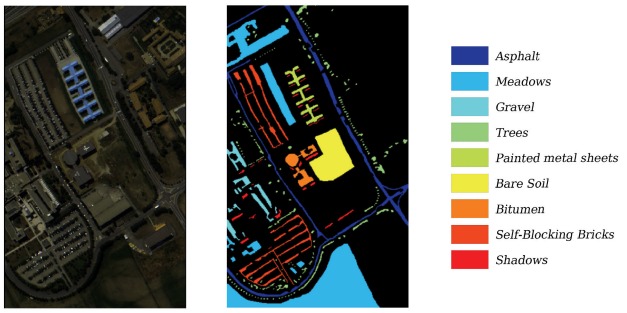



Fig. 9. (a) False-color composition of the Pavia University scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes.

TABLE III

NUMBER OF TRAINING, TESTING, AND TOTAL SAMPLES FOR EACH LAND-COVER TYPE IN THE PAVIA UNIVERSITY DATASET

| Class No. | Land Cover Type | Train Num | Test Num | Total Num |
|---|---|---|---|---|
| C1 | Asphalt | 100 | 6531 | 6631 |
| C2 | Meadows | 100 | 18549 | 18649 |
| C3 | Gravel | 100 | 1999 | 2099 |
| C4 | Trees | 100 | 2964 | 3064 |
| C5 | Painted metal sheets | 100 | 1245 | 1345 |
| C6 | Bare Soil | 100 | 4929 | 5029 |
| C7 | Bitumen | 100 | 1230 | 1330 |
| C8 | Self-Blocking Bricks | 100 | 3582 | 3682 |
| C9 | Shadows | 100 | 847 | 947 |
| | Total | 900 | 41876 | 42776 |

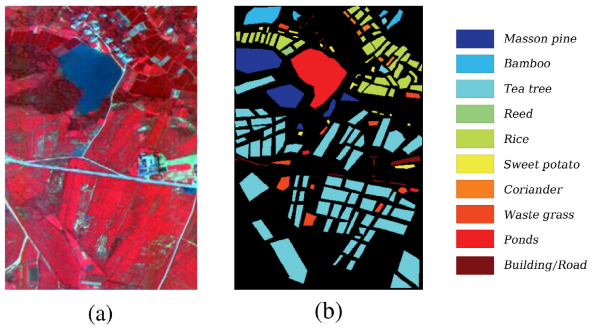



Fig. 10. (a) False-color composition of the Fanglu Tea Farm scene. (b) Ground-truth map containing ten mutually exclusive land-cover classes.

the tea planting base in Fanglu Village, Changzhou City, Jiangsu Province, China. The data contain 348 × 512 pixels, with a spatial resolution of 2.25 m and 80 spectral bands covering a spectral range of 417–855 nm. Fig. 10 shows a false-color composite of the image and the corresponding ground-truth map. The dataset contains 53 734 labeled pixels and ten land-cover classes of interest. The training and testing divisions used in the experiments are listed in Table IV.

### B. Analysis on Different Groups and Layers

Obviously, according to (3), we find out that the larger the group number $g$ of the grouped convolution, the less the number of parameters of the model. So in this experiment, we analyzed and compared the performance of different grouping schemes. We set the number of grouped convolution $g$ (used to divide channels) to 1/2/4/10, respectively.

TABLE IV

NUMBER OF TRAINING, TESTING, AND TOTAL SAMPLES FOR EACH LAND-COVER TYPE IN THE FANGLU TEA FARM DATASET

| Class No. | Land Cover Type | Train Num | Test Num | Total Num |
|---|---|---|---|---|
| C1 | Masson pine | 100 | 5706 | 5806 |
| C2 | Bamboo forest | 100 | 2218 | 2318 |
| C3 | Tea tree | 100 | 28328 | 28428 |
| C4 | Reed | 100 | 114 | 214 |
| C5 | Rice | 100 | 6709 | 6809 |
| C6 | Sweet potato | 100 | 717 | 817 |
| C7 | Coriander | 100 | 329 | 429 |
| C8 | Waste grass | 100 | 1761 | 1861 |
| C9 | Ponds | 100 | 811 | 911 |
| C10 | Building/road | 100 | 847 | 947 |
| | Total | 1000 | 52734 | 53734 |

TABLE V

DIFFERENT GROUPED NUMBERS, AND DIFFERENT GROUPED CONVOLUTION SCHEMES THAT PERFORMED ON THE INDIAN PINES DATASET

| | | First 1 layer | First 2 layers | First 3 layers | First 4 layers |
|---|---|---|---|---|---|
| g=1 | Parameters<br>Time/Epoch<br>OA | | | | **2,588,656**<br>**0.21s**<br>**98.64%** |
| g=2 | Parameters<br>Time/Epoch<br>OA | 2,540,656<br>0.22s<br>**98.73%** | 2,451,865<br>0.22s<br>98.72% | 2,157,459<br>0.21s<br>98.66% | **979,856**<br>**0.20s**<br>98.62% |
| g=4 | Parameters<br>Time/Epoch<br>OA | 2,526,256<br>0.23s<br>**98.73%** | 2,405,056<br>0.22s<br>98.57% | 2,053,056<br>0.22s<br>98.34% | **645,056**<br>**0.20s**<br>98.29% |
| g=10 | Parameters<br>Time/Epoch<br>OA | 2,517,616<br>0.25s<br>**98.57%** | 2,376,976<br>0.25s<br>98.47% | 1,990,416<br>0.25s<br>98.19% | **444,176**<br>**0.23s**<br>97.94% |

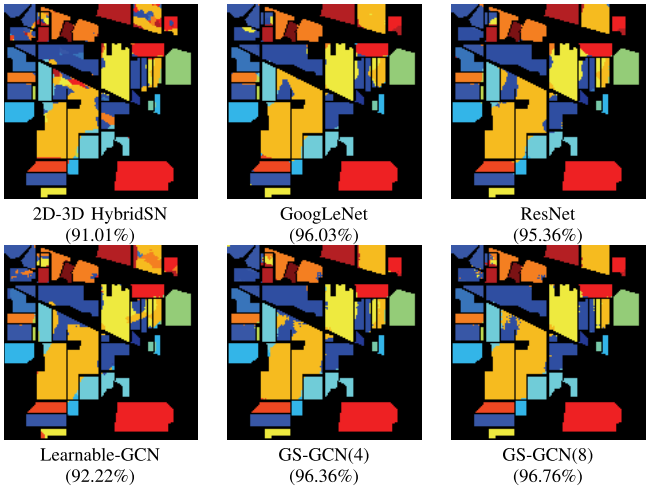



Fig. 11. Classification maps (along with the OAs) obtained by different models for the Indian Pines dataset. These maps correspond to a single experiment in Table VIII.

In addition, we also analyzed the efficiency of GSC when it is implemented on the first 1/2/3/4 layers of the network, which means the remaining convolution layers are all GCN layers without being grouped. Here, we set the orientations of Gabor filters to $M = 4$, and the number of output channels of each layer is 4 × 60/80/160/320 (in Section III-C, we will discuss the model size involving the number of channels and the number of GMFs' orientations in detail). We choose the Indian Pines dataset in this experiment, and the training samples account for 10% of the total labeled samples.

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TABLE VI
PERFORMANCE ON FOUR DATASETS, OF WHICH ALL SIX MODELS CONTAIN 80/160/240/320 CHANNELS IN EACH LAYER

| Dataset | Model | $\kappa$ | AA(%) | OA(%) | Parameter Size | Memory(MB) | Time(ms) |
|---|---|---|---|---|---|---|---|
| Indian Pines | CNN | 0.91 | 94.71 | 93.74 | 1,173,136 | 4.48 | 0.057 |
| | GS-CNN | 0.92 | 94.89 | 93.70 | 541,336 | 2.07 | **0.056** |
| | GCN(4) | 0.95 | 95.78 | 95.18 | 309,136 | 1.18 | 0.066 |
| | GS-GCN(4) | **0.96** | **97.41** | 95.09 | 148,336 | 0.57 | 0.058 |
| | GCN(8) | 0.96 | 97.12 | **95.26** | 165,136 | 0.63 | 0.066 |
| | GS-GCN(8) | 0.95 | 96.12 | 94.97 | **89,136** | **0.34** | 0.079 |
| Salinas | CNN | 0.98 | 97.65 | 97.56 | 1,173,136 | 4.48 | 0.100 |
| | GS-CNN | 0.97 | 97.61 | 97.52 | 541,336 | 2.07 | **0.059** |
| | GCN(4) | 0.98 | 98.79 | 98.61 | 309,136 | 1.18 | 0.084 |
| | GS-GCN(4) | 0.98 | 98.91 | 98.70 | 148,336 | 0.57 | 0.061 |
| | GCN(8) | 0.99 | 99.01 | 98.87 | 165,136 | 0.63 | 0.074 |
| | GS-GCN(8) | **0.99** | **99.12** | **99.01** | **89,136** | **0.34** | 0.082 |
| Pavia University | CNN | 0.95 | 96.12 | 96.07 | 1,173,136 | 4.48 | 0.057 |
| | GS-CNN | 0.96 | 96.03 | 96.00 | 536,939 | 2.05 | **0.042** |
| | GCN(4) | 0.97 | 97.62 | 97.41 | 299,689 | 1.14 | 0.046 |
| | GS-GCN(4) | 0.97 | 97.60 | 97.33 | 139,889 | 0.53 | 0.044 |
| | GCN(8) | **0.98** | **98.01** | **97.68** | 155,689 | 0.59 | 0.046 |
| | GS-GCN(8) | 0.97 | 97.81 | 97.41 | **75,289** | **0.29** | 0.050 |
| Fanglu Tea Farm | CNN | 0.97 | 97.43 | 97.31 | 1,173,136 | 4.48 | 0.045 |
| | GS-CNN | 0.97 | 97.24 | 97.38 | 537,260 | 2.05 | **0.042** |
| | GCN(4) | 0.98 | 98.53 | 98.42 | 300,010 | 1.14 | 0.046 |
| | GS-GCN(4) | 0.98 | 98.56 | 98.44 | 140,210 | 0.53 | 0.043 |
| | GCN(8) | 0.98 | 98.62 | **98.55** | 156,010 | 0.60 | 0.049 |
| | GS-GCN(8) | **0.98** | **98.64** | 98.50 | **75,610** | **0.29** | 0.050 |

TABLE VII
PERFORMANCE ON FOUR DATASETS, OF WHICH ALL SIX MODELS CONTAIN 240/320/640/1280 CHANNELS IN EACH LAYER

| Dataset | Model | $\kappa$ | AA(%) | OA(%) | Parameter Size | Memory(MB) | Time(ms) |
|---|---|---|---|---|---|---|---|
| Indian Pines | CNN | 0.95 | 95.12 | 94.95 | 9,975,856 | 38.05 | 0.107 |
| | GS-CNN | 0.95 | 95.01 | 94.89 | 3,696,056 | 14.1 | **0.092** |
| | GCN(4) | 0.96 | 97.09 | 96.59 | 2,545,456 | 9.71 | 0.107 |
| | GS-GCN(4) | **0.96** | 97.41 | 96.36 | 965,456 | 3.68 | 0.102 |
| | GCN(8) | 0.96 | 97.15 | 96.32 | 1,307,056 | 4.99 | 0.119 |
| | GS-GCN(8) | 0.96 | **98.23** | **96.76** | **502,656** | **1.92** | 0.108 |
| Salinas | CNN | 0.97 | 98.15 | 97.81 | 9,975,856 | 38.05 | 0.106 |
| | GS-CNN | 0.98 | 98.20 | 97.85 | 3,696,056 | 14.1 | **0.086** |
| | GCN(4) | 0.99 | 99.32 | 99.00 | 2,545,456 | 9.71 | 0.107 |
| | GS-GCN(4) | **0.99** | **99.64** | **99.09** | 965,456 | 3.68 | 0.104 |
| | GCN(8) | 0.99 | 99.23 | 98.98 | 1,307,056 | 4.99 | 0.117 |
| | GS-GCN(8) | 0.99 | 99.51 | 99.04 | **502,656** | **1.92** | 0.111 |
| Pavia University | CNN | 0.98 | 98.37 | 98.02 | 9,945,289 | 37.94 | 0.092 |
| | GS-CNN | 0.98 | 98.29 | 98.01 | 3,696,056 | 14.1 | **0.070** |
| | GCN(4) | 0.98 | **99.14** | **98.77** | 2,514,889 | 9.59 | 0.085 |
| | GS-GCN(4) | **0.98** | 98.30 | 98.59 | 949,289 | 3.62 | 0.076 |
| | GCN(8) | 0.98 | 98.86 | 98.66 | 1,276,489 | 4.87 | 0.092 |
| | GS-GCN(8) | 0.98 | 98.06 | 98.45 | **480,489** | **1.83** | 0.078 |
| Fanglu Tea Farm | CNN | 0.97 | 97.68 | 97.45 | 9,946,570 | 37.94 | 0.079 |
| | GS-CNN | 0.97 | 97.69 | 97.49 | 3,684,620 | 14.06 | **0.067** |
| | GCN(4) | 0.98 | 98.77 | 98.76 | 2,516,170 | 9.6 | 0.091 |
| | GS-GCN(4) | 0.98 | 99.12 | 98.89 | 950,570 | 3.63 | 0.078 |
| | GCN(8) | 0.98 | 99.16 | **98.96** | 1,277,770 | 4.87 | 0.091 |
| | GS-GCN(8) | **0.98** | **99.41** | 98.91 | **481,770** | **1.84** | 0.078 |

The comparison results are shown in Table V. Among, the row of $g = 1$ means no GSC, which is denoted as the basic GCN. From the perspective of grouping, when the number of GSC layers is the same, the larger the number of groups $g$, the smaller the number of parameters, while the longer the reasoning time and the lower the classification accuracy. It is because the process of grouped operation takes more times. At the same time, the more dispersed the groups, the lower the channel codependence, resulting in lower classification accuracy. On the other side, from the perspective of applying GSC on different number of layers, the more grouped separable layers applied, the smaller the number of parameters, and the shorter the reasoning time, but the accuracy also decreases.

Note that the number of channels of feature maps in our model almost doubles with the deepening of the number of layers. Therefore, with the deepening of applying the number of GSC layers, the parameters of GS-GCN model will decrease exponentially. By comparison, when $g = 2$ and 4 and GSC is applied to the first layer, we find that its accuracy (98.73%) is even higher than that of the basic GCN (98.64%). When $g = 2$ and GSC is applied to the first four layers,

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TABLE VIII
OAs[%], AAs[%], AND $\kappa$[%] STATISTIC FOR THE INDIAN PINES SCENE

| Class | 2D-3D HybridSN | GoogLeNet | ResNet | Learnable-GCN | GS-GCN(4) | GS-GCN(8) |
|---|---|---|---|---|---|---|
| C1 | 76.92 | 83.33 | 100.00 | 90.00 | 90.00 | **100.00** |
| C2 | 88.77 | 93.14 | 92.10 | 87.92 | **94.33** | 91.41 |
| C3 | 88.98 | 94.11 | 94.66 | 88.22 | **96.33** | 95.67 |
| C4 | 84.33 | 100.00 | 91.96 | 96.67 | **100.00** | 99.47 |
| C5 | 99.27 | 96.44 | 96.71 | 97.37 | **99.76** | 99.52 |
| C6 | 88.90 | 96.68 | 93.22 | 93.15 | 96.30 | **97.54** |
| C7 | 100.00 | 46.15 | 66.67 | 54.55 | **100.00** | 100.00 |
| C8 | 98.17 | 99.07 | 99.76 | 99.53 | **100.00** | 99.77 |
| C9 | 50.00 | 28.57 | 66.67 | 20.00 | **100.00** | 66.67 |
| C10 | **97.84** | 94.95 | 88.95 | 92.00 | 97.00 | 96.75 |
| C11 | 92.62 | **98.87** | 98.67 | 91.25 | 95.57 | 98.82 |
| C12 | 81.89 | 91.55 | 92.39 | 89.89 | **95.14** | 93.30 |
| C13 | 78.28 | 91.18 | **100.00** | 95.09 | 99.36 | 99.36 |
| C14 | 92.77 | 100.00 | 99.06 | **100.00** | 99.92 | 99.17 |
| C15 | 88.19 | 90.71 | **97.38** | 87.66 | 87.96 | 95.18 |
| C16 | 91.30 | 95.35 | **97.67** | 91.49 | 91.49 | 91.30 |
| OA(%) | 91.04 | 96.03 | 95.36 | 92.22 | 96.36 | **96.76** |
| AA(%) | 95.97 | 97.59 | 97.49 | 95.20 | 97.41 | **98.23** |
| $\kappa$ | 0.8976 | 0.9544 | 0.9468 | 0.9106 | 0.9581 | **0.9627** |

TABLE IX
OAs[%], AAs[%], AND $\kappa$[%] STATISTIC FOR THE SALINAS SCENE

| Class | 2D-3D HybridSN | GoogLeNet | ResNet | Learnable-GCN | GS-GCN(4) | GS-GCN(8) |
|---|---|---|---|---|---|---|
| C1 | 100.00 | 100.00 | 100.00 | 100.00 | **100.00** | **100.00** |
| C2 | 98.92 | 99.92 | 99.92 | 100.00 | **100.00** | 100.00 |
| C3 | 100.00 | 98.77 | 100.00 | 96.25 | **100.00** | 100.00 |
| C4 | **98.85** | 98.44 | 95.45 | 92.02 | 97.88 | 96.20 |
| C5 | 99.81 | 99.43 | 99.88 | 99.52 | 99.81 | **99.85** |
| C6 | 100.00 | 99.97 | 100.00 | 100.00 | **100.00** | 100.00 |
| C7 | 98.94 | 97.11 | 100.00 | 99.69 | 99.89 | **100.00** |
| C8 | 99.61 | 98.89 | **99.06** | 98.14 | 99.61 | 99.30 |
| C9 | 100.00 | 100.00 | 100.00 | 99.77 | **100.00** | 100.00 |
| C10 | 99.50 | 99.69 | 99.91 | 98.29 | **100.00** | 99.91 |
| C11 | 96.40 | 98.36 | 100.00 | 99.03 | **100.00** | 97.42 |
| C12 | 99.84 | 94.64 | 98.51 | 98.84 | **100.00** | 100.00 |
| C13 | 98.54 | 99.64 | 96.98 | 99.88 | **100.00** | 99.88 |
| C14 | 99.51 | 99.02 | 99.32 | 99.61 | **100.00** | 99.03 |
| C15 | 98.49 | 96.66 | **89.92** | 83.11 | 94.63 | 95.47 |
| C16 | 100.00 | 99.66 | 100.00 | 99.94 | **100.00** | 100.00 |
| OA(%) | 98.57 | 98.71 | 95.36 | 96.45 | **99.09** | 99.04 |
| AA(%) | 98.72 | 98.56 | 97.49 | 98.36 | **99.64** | 99.51 |
| $\kappa$ | 0.9852 | 0.9856 | 0.9468 | 0.9605 | **0.9898** | 0.9893 |

TABLE X
OAs[%], AAs[%], AND $\kappa$[%] STATISTIC FOR THE PAVIA UNIVERSITY SCENE

| Class | 2D-3D HybridSN | GoogLeNet | ResNet | Learnable-GCN | GS-GCN(4) | GS-GCN(8) |
|---|---|---|---|---|---|---|
| C1 | 98.22 | 98.14 | 98.92 | 96.21 | 99.30 | **99.36** |
| C2 | **99.71** | 99.64 | 99.50 | 97.51 | 99.63 | 99.56 |
| C3 | 96.41 | 96.75 | 94.86 | 84.46 | **97.68** | 94.61 |
| C4 | 91.58 | 96.44 | **97.89** | 91.46 | 95.12 | 95.94 |
| C5 | **100.00** | 99.36 | 99.68 | 99.36 | 98.34 | 97.49 |
| C6 | 99.80 | 99.84 | 99.92 | 88.08 | 99.82 | **99.90** |
| C7 | **98.40** | 84.45 | 84.74 | 87.59 | 91.01 | 93.80 |
| C8 | 93.07 | 96.00 | 94.38 | 82.94 | **96.95** | 96.25 |
| C9 | 95.32 | 97.01 | **98.31** | 79.27 | 97.32 | 95.19 |
| OA(%) | 98.03 | 98.18 | 98.14 | 93.07 | **98.59** | 98.45 |
| AA(%) | 97.93 | 97.75 | 97.67 | 92.58 | **98.30** | 98.06 |
| $\kappa$(%) | 0.9738 | 0.9757 | 0.9752 | 0.9079 | **0.9812** | 0.9793 |

its parameters' number (979 856) accounts for about 3/8 of the basic GCN (2 588 656), and its accuracy still maintains 98.62%, only 0.02% lower than the basic GCN. Compared with larger groups, when the number of GSC layers becomes deeper, the number of parameters decreases significantly, but the accuracy cannot be maintained, and the reasoning time will become longer.

According to the above experimental results, and considering that our goal is to minimize the parameter quantity as much as possible except for maintaining or even exceeding the

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TABLE XI
OAs[%], AAs[%], AND $\kappa$[%] STATISTIC FOR THE FANGLU TEA FARM SCENE

| Class | 2D-3D HybridSN | GoogLeNet | ResNet | Learnable-GCN | GS-GCN(4) | GS-GCN(8) |
|---|---|---|---|---|---|---|
| C1 | 98.89 | 99.77 | 98.94 | 96.21 | 98.53 | **99.89** |
| C2 | 88.99 | 89.04 | 82.26 | 61.59 | **99.91** | 95.23 |
| C3 | 99.79 | 99.93 | 99.81 | 99.29 | 99.87 | **99.98** |
| C4 | 81.43 | 63.33 | 67.86 | 55.88 | **100.00** | 97.44 |
| C5 | 98.80 | 99.97 | 99.78 | 98.17 | **99.97** | 99.90 |
| C6 | 92.77 | 84.52 | 86.16 | 87.94 | **98.88** | 92.86 |
| C7 | 97.34 | 74.60 | 67.42 | 66.73 | **99.70** | 80.05 |
| C8 | 72.25 | 79.33 | 76.04 | 48.26 | 79.88 | **87.05** |
| C9 | 100.00 | 100.00 | 100.00 | 100.00 | **100.00** | 100.00 |
| C10 | 85.71 | 95.89 | 98.14 | **99.10** | 98.40 | 94.93 |
| OA(%) | 97.41 | 97.91 | 97.18 | 92.91 | 98.89 | **98.91** |
| AA(%) | 98.89 | 98.96 | 98.59 | 95.01 | 99.12 | **99.41** |
| $\kappa$(%) | 0.9618 | 0.9690 | 0.9584 | 0.8974 | 0.9835 | **0.9838** |

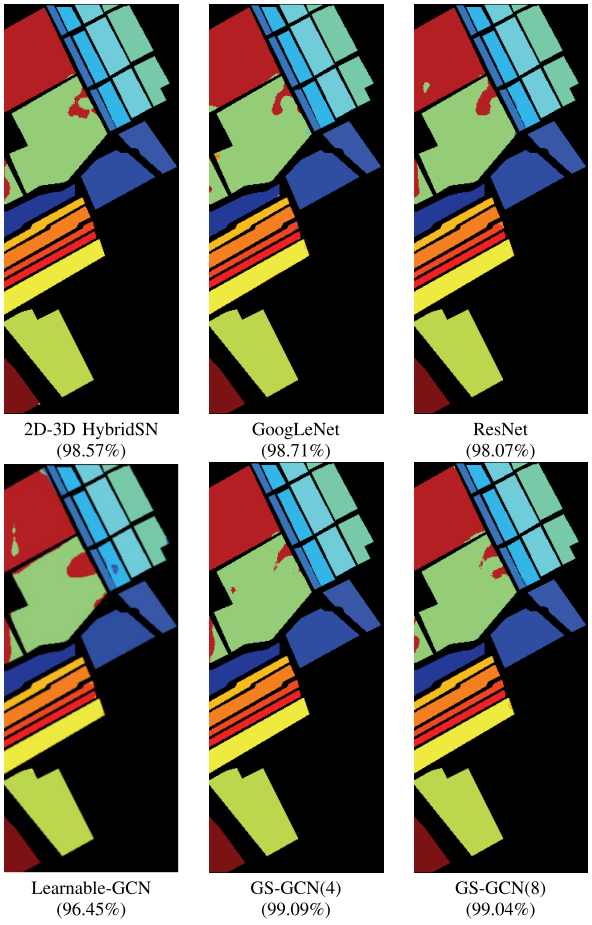



Fig. 12. Classification maps (along with the OAs) obtained by different models for the Salinas dataset. These maps correspond to a single experiment in Table IX.

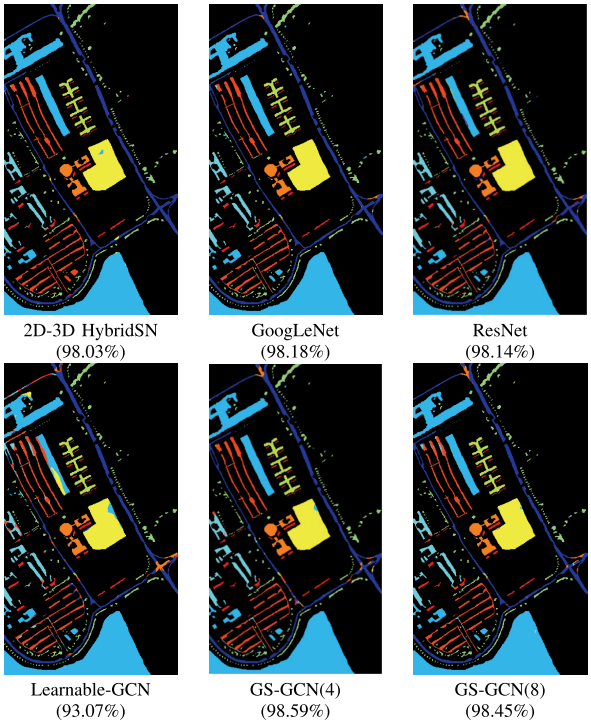



Fig. 13. Classification maps (along with the OAs) obtained by different models for the Pavia University dataset. These maps correspond to a single experiment in Table X.

accuracy of the original GCN, we suggest to perform GSC in all the layers, as it has the smallest parameter quantity and only a slight difference in reasoning time. In addition, by choosing an appropriate group number ($g = 2$ in our experiments), more higher accuracy can be ensured.

In addition, since multiscale convolution is commonly used in many popular models and brings effective results, we further explored various grouping strategies from a deeper perspective and carried out experiment on the multiscale grouping strategy. We allocated different grouping schemes to distinct branches. Concretely, a total of four branches with group sizes of one, two, four, and eight are adopted. Then, feature maps are processed using multilevel grouped separable Gabor convolutions across different branches. Finally, the resulting feature maps of all the branches are aggregated for final classification.

However, the experimental results are not as good as we have hoped, and the implementation of multiple branches (here, each branch represents a different grouping scheme) results in a significant increase in parameters' quantity and reasoning time, while just a slight accuracy improvement. As a result, from the point of balancing accuracy and computational efficiency, we do not adopt multiscale scheme.

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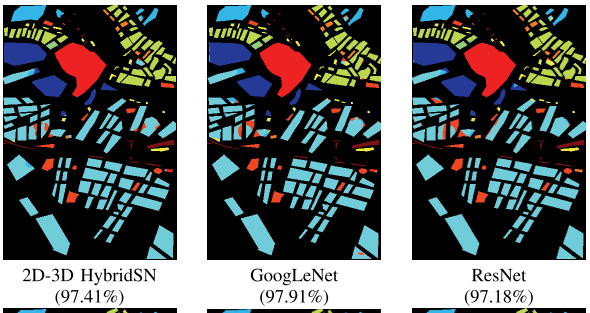



Fig. 14. Classification maps (along with the OAs) obtained by different models for the Fanglu Tea Farm dataset. These maps correspond to a single experiment in Table XI.

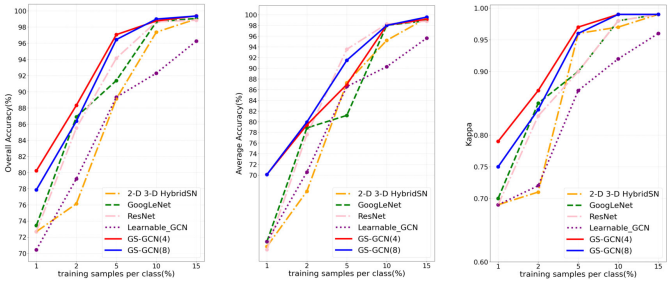



Fig. 15. Classification accuracies versus different percentages of training sample per class on the Indian Pines dataset. (a) OA(%). (b) AA(%). (c) Kappa.

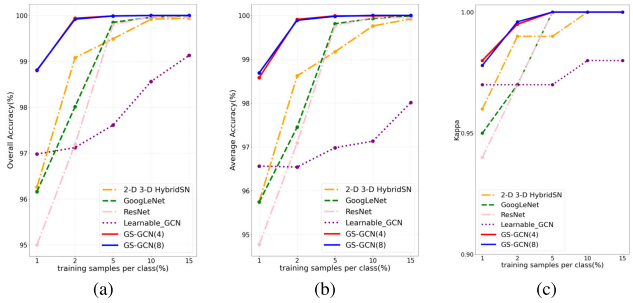



Fig. 16. Classification accuracies versus different percentages of training sample per class on the Salinas dataset. (a) OA(%). (b) AA(%). (c) Kappa.

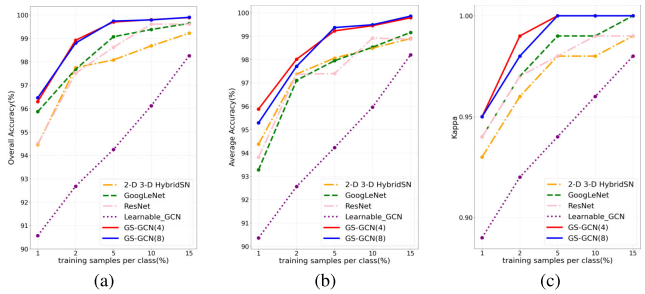



Fig. 17. Classification accuracies versus different percentages of training sample per class on the Pavia University dataset. (a) OA(%). (b) AA(%). (c) Kappa.

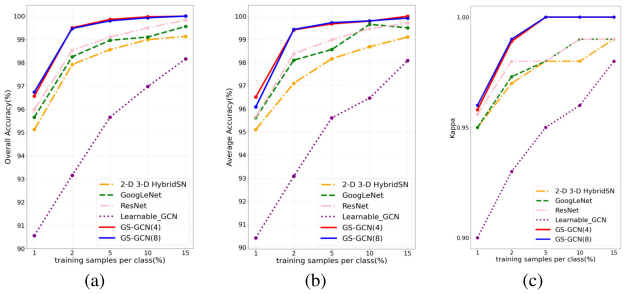



Fig. 18. Classification accuracies versus different percentages of training sample per class on the Fanglu Tea Farm dataset. (a) OA(%). (b) AA(%). (c) Kappa.

TABLE XII

PERFORMANCE OF SIX MODELS ON FOUR DATASETS

| Data set | Model | Parameter Size | Memory (MB) | Time (ms) |
|---|---|---|---|---|
| Indian Pines | 2D-3D HybridSN | 5,122,176 | 19.54 | 0.178 |
| | GoogLeNet | 1,932,184 | 7.37 | **0.116** |
| | ResNet | 2,051,456 | 7.83 | 0.128 |
| | Learnable GCN | 3,105,232 | 11.85 | 0.366 |
| | GS-GCN(4) | 965,456 | 3.68 | 0.121 |
| | GS-GCN(8) | **502,656** | **1.92** | 0.121 |
| Salinas | 2D-3D HybridSN | 5,122,176 | 19.54 | 0.213 |
| | GoogLeNet | 1,932,184 | 7.37 | 0.111 |
| | ResNet | 2,051,456 | 7.83 | **0.105** |
| | Learnable GCN | 3,105,232 | 11.85 | 0.432 |
| | GS-GCN(4) | 965,456 | 3.68 | 0.104 |
| | GS-GCN(8) | **502,656** | **1.92** | 0.111 |
| Pavia University | 2D-3D HybridSN | 5,121,273 | 19.54 | 0.216 |
| | GoogLeNet | 1,832,577 | 6.99 | 0.080 |
| | ResNet | 2,035,865 | 7.77 | **0.075** |
| | Learnable GCN | 2,984,329 | 11.38 | 0.332 |
| | GS-GCN(4) | 949,289 | 3.62 | 0.076 |
| | GS-GCN(8) | **480,489** | **1.83** | 0.078 |
| Fanglu Tea Farm | 2D 3D HybridSN | 5,121,402 | 19.54 | 0.212 |
| | GoogLeNet | 1,832,577 | 6.99 | 0.081 |
| | ResNet | 2,036,378 | 7.77 | **0.072** |
| | Learnable GCN | 2,984,329 | 11.38 | 0.241 |
| | GS-GCN(4) | 950,570 | 3.63 | 0.078 |
| | GS-GCN(8) | **481,770** | **1.84** | 0.078 |

### *C. Analysis on Different Numbers of Channels and Orientations*

In this section, we discussed the impact of the number of output channels and orientations in Gabor filters in each layer. We designed two experiments and compared our GS-GCN with the models in Fig. 1. Here, the number of orientations of Gabor filters is set to $M = 4, 8$, and we call them as GCN(4), GCN(8), GS-GCN(4), and GS-GCN(8). While for the channel configuration of all the models. two types of settings are adopted and represented as 80/160/240/320 and 240/320/640/1280, respectively, which means there are four layers, and the number of channels increases layer by layer. In addition, for the Gabor scale parameters at different layers, we configured with scales descending from coarse to fine. Specifically, the scale parameter starts at 1, and then gradually decreases by dividing the square root of 2. We also carried

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out parallel experiments on the setting of scale parameters ascending from fine to coarse, but the results are not as effective as the coarse-to-fine setup. We also compared four similar models with the same number of channels: standard CNN, GS-CNN with only GSC, GCN with only GMFs, and our GS-GCN.

Tables VI and VII show the model performance of 80/160/240/320 channels and 240/320/640/1280 channels, respectively. Six indicators are compared, among, $\kappa$ stands for the Kappa coefficient of the classification results, **AA** stands for the average accuracy of multiple runs, **OA** stands for the overall accuracy, **Parameter Size** refers to the convolution kernel parameters of the network model, **Memory** is the video memory allocation of the network model, and **Time** is the average reasoning time for each sample.

As observed from Tables VI and VII, the larger the number of channels, the greater the parameter quantity and reasoning time. At the same time, the accuracy rate also increased, but not very obvious, about 0.5%–1.5%. For different models, the largest model parameter is owned by standard CNN, and the smallest is the eight-orientational GS-GCN(8) (the more the number of orientations of Gabor filters, the smaller the model parameters). The parameters of the latter are nearly 1/10 of those of the former, but the accuracy is better than the standard CNN and is basically the same as that of the GCN, about ±0.02%–0.3%. However, as the number of orientations of GMFs increases, the reasoning time will increase accordingly. The reasoning time of the four-orientational GS-GCN(4) network is slightly larger than that of the grouped separable GS-CNN network (the fastest model), ranking second in all the models. However, the reasoning time of the eight-orientational GS-GCN network will be much slower, even slower than the standard CNN.

### D. Classification Comparison With State-of-the-Art Models

To further demonstrate the performance superiority of our GS-GCN model, we compare with some classic CNN-based classification models. The models involved in the comparison are listed as follows.

1) *2-D–3-D HybridSN:* This is a mixed 2-D and 3-D CNN model, including three 3-D convolution layers, one 2-D convolution layer, and three full connection layers.
2) *GoogLeNet:* Here, we modified the structure of GoogLeNet, which has four convolution layers and a full connection layer. The convolution layer is 2-D CNN and inception module in turn.
3) *ResNet:* Here, we simplified the network structure of ResNet, which is mainly composed of 2-D CNN + ResBlock + 2-D CNN + ResBlock + Linear1 + Linear2.
4) *Learnable-GCN:* In this model, the parameters of Gabor filters are learnable, but the training process is time-consuming, so we only replace the first layer of the model with 2-D Gabor CNN, the next two convolution layers still use 2-D CNN, and finally use two fully connected layers for output.

We set the channel number of our model as 240/320/640/1280 and use the four-orientation and eight-orientation models for experimental comparison. In terms of performance quantitative indicators, the experiment adopted overall accuracy (OA), average accuracy (AA), and Kappa coefficient of the classification results ($\kappa$). To be exact, OA is the number of correctly classified samples divided by the total number of test samples, and AA is the average of the classification accuracy of all land-cover types. In addition, Kappa coefficient is a statistical measurement, especially in multiclassification tasks with imbalanced samples.

The classification results are shown in Tables VIII–XI, respectively. From the classification results, we find that the GS-GCN model has very good classification performance with small parameters. In each dataset, whether four-orientation or eight-orientation models, the accuracies are higher than other models. In the Indian Pines dataset, our eight-orientation GS-GCN(8) has the best OA, AA, and $\kappa$, the four-orientation GS-GCN(4) has the second highest performance, and the classification accuracy of many categories ranks first. In contrast, the classification accuracy of other models still lags behind the GS-GCN model. For the Salinas dataset, it is worth noting that the samples we used for training are less than 1.5% of the total samples. Our four-orientation GS-GCN(4) model has the best overall performance, its OA reached 99.09%, nearly 0.5% higher than 2-D–3-D HybridSN model, and nearly 3.5% higher than ResNet. It is because the GS-GCN model can extract features in multiple orientations and reduce the mutual coupling of convolution kernels, thus more robust and adaptable to small training samples. In the Pavia University dataset, four-orientation GS-GCN(4) has the best overall performance, followed by GS-GCN(8). Their OA is more than 0.3% higher than that of GoogLeNet. For the Fanglu Tea Farm dataset, the number of training samples we used is less than 2% of the total samples, and the OA of our eight-orientation GS-GCN(8) is 98.91%, with 1% higher than that of GoogLeNet.

Finally, for illustrative purposes, Figs. 11–14, respectively, present the classification maps obtained for the considered images by different models. As we can observe, noise disturbance and misclassification phenomena still exist in the classification maps obtained using 2-D–3-D HybridSN, especially in the high mixed Indian pines data, e.g., c1-Alfalfa, c9-Oats, and c13-Wheat, which have 36, 16, and 50 training samples, respectively, and the classification accuracies are 76.92%, 50%, and 78.28%, respectively. While for the GS-GCN(4) model, nearly 100% accuracies are obtained, which further denotes that the GS-GCN model can make full use of multiscale and multiorientational features to reflect spatial distribution details of different scales ground objects, thus leading to a better classification performance.

### E. Experiments With Varying Proportion of Training Samples

To further assess the performance of our proposed GS-GCN, we conducted experiments on different proportion of training samples. Specifically, we tested five values (1%, 2%, 5%, 10%, and 15%) to represent the proportion of randomly selected training samples from each class and then the remaining samples for testing. In addition, the above-mentioned four

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classic models are also used for comparison. We take OA, AA, and Kappa as indicators, and the corresponding trend curves are shown in Figs. 15–18. From the experimental results, we might find that the performance of our proposed GS-GCN is quite excellent compared with other models, especially when the number of training samples is small (e.g., 1%, 2%), which still keeps a higher classification accuracy. At the same time, it can be ensured that the classification accuracies are also ahead of other models when the training samples become larger (e.g., 5%, 10%, 15%).

### F. Analysis on Computational Complexity

In addition, we compared the number of parameters, memory allocation, and reasoning time of the four-orientation and eight-orientation GS-GCN models with the above four classic CNN-based models in Table XII. It can be seen that the parameters of two GS-GCN models are far less than those of other models, especially the eight-orientation model. In terms of reasoning time, although GS-GCN model does not have outstanding performance, the reasoning time is close to the minimum one of all the models. We believe that the reason is that the GS-GCN model takes a certain amount of time to perform grouped convolution and generate multiorientational Gabor filters. The main purpose of introducing GS-GCN is to greatly simplify the network parameters and make training easier. From the effect point of view, the combination of GCN and GSC can maintain good classification accuracy, while significantly reducing the number of parameters.

## IV. Conclusion and Future Research

In this article, we developed a new lightweight network model, called GS-GCN, for advanced land-cover classification of remotely sensed HSIs. Our model is composed of four convolution layers and a full connection layer. For each convolution layer, we use GMFs that can generate multidirectional features to replace the traditional CNN filter. At the same time, we apply grouping separable convolutions to greatly reduce the coupling between convolution cores and the codependence within the network, thus greatly reducing the network parameters and effectively avoiding overfitting problem. In the experiment, we first compared our different group schemes and selected the one with better performance as the main structure of our model. We compared our model with other similar models and other state-of-the-art classification models in four open datasets. From the experimental results, we can see that our model can maintain excellent classification accuracy on the premise of reducing a large number of parameters. It should be noted that the GS-GCN model does not exhibit superior performance in terms of reasoning time. This is because the modulation process of Gabor convolution consumes a considerable amount of time, and this challenge paves the way for our future research direction. In the future, we will aim to optimize the Gabor filter function and explore more lightweight and more suitable multiscale and multiorientational feature expressions for HSIC.

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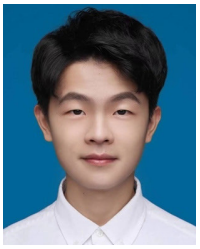
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